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PI Name:	Baulch, Janet E. Ph.D.		
Project Title:	Space Radiation Effects on Genomic and Epigenetic Stability: Influence of Radiation Quality and Genetic Predisposition		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Comments:	NOTE: PI previously at University of Maryland; moved to UCA Irvine in summer 2013 (Ed., 11/11/13)		
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Task Description:	[Editor's note 11/11/2013. Note this is a continuation of "Space Radiation Effects on Genomic and Epigenetic Stability: Influence of Radiation Quality and Genetic Predisposition" (grant NNX11AC55G); see that project for previous reports.] Astronauts traveling in space are exposed to a variety of radiations. It is known that exposure to ionizing radiation can lead to an increased risk for developing cancer. Since there is currently no way to protect astronauts, it is important that the carcinogenic risk of exposures to these radiations is understood. The delayed, deleterious effects of radiation exposure are grouped under the terms radiation-induced genomic and epigenomic instability. These mechanisms are considered to be early steps in the carcinogenic process. Epigenetic mechanisms are thought to play a significant role in carcinogenesis by regulating gene expression and genome stability. These epigenetic mechanisms include DNA methylation, miRNA mediated gene silencing, and histone modification. There are also genetic factors in mouse and		

	humans that determine the relative sensitivity of their cells to radiation-induced effects and ultimately their cancer risk. This project will develop assays that will provide functional biomarkers for exposure that are also indicators of carcinogenic risk. We will accomplish these objectives by evaluating mice with differing sensitivities to radiation for epigenomic instability as measured by changes in miRNA and DNA methylation profiles following exposure to iron, silicon, or oxygen ion radiations. The effect of age on the epigenetic radiation induced response will be evaluated using mice of relatively younger and older ages.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The goal of this grant is to evaluate the effect of space radiation exposure on epigenetic profiles with a focus on the role for genetic factors, age, and radiation linear energy transfer (LET). Some of the research and technology questions that this proposal addresses include the following: 1) the probabilities for increased carcinogenic risk following relatively low dose radiation exposure as a function of NASA's operational parameters (radiation quality, age at exposure, genetic factors, tissue specific response); 2) the development of quantitative, functional biomarkers for carcinogenic risk; 3) a better understanding of the molecular mechanisms involved in cancer initiation and progression. Understanding the cancer risk associated with exposure to either terrestrial and space radiation is important for human health. Epigenetic abnormalities are thought to play a key role in the earliest steps of carcinogenic initiation. Epigenetic changes are mitotically and meiotically heritable, stable alteration in gene expression that include DNA methylation, histone modifications, and RNA associated gene silencing. This project tests the hypothesis that radiation exposure can affect epigenetic profiles and increase carcinogenic risk. We further hypothesize that genetic factors and age at the time of exposure impact the epigenetic radiation response. Specific Aim 1 will evaluate radiation-induced alterations in miR profiles and gene expression. For each tissue, dose, and time point, genome-wide miR profiles will be analyzed. In parallel, global mRNA profiles will be analyzed to provide information regarding gene expression. Together these arrays will allow us to identify up-regulated miRNA that correlate with down-regulated target mRNA, or vis versa. Mining these data will allow us to identify affected pathways that are relevant to cancer and the radiation response. Specific aim 2 will evaluate the effect of irradiation on the induction of aberrant DNA methylation. Alterations in DNA methylation have emerged as one of the most consistent molecular markers in multiple cancers and for carcinogenic initiation and progression. We will measure genome wide methylation by evaluating repeat element methylation. These repeat elements make up a significant portion of the mammalian genome and aberrant repeat element methylation has been linked to genomic instability, altered gene expression and cancer. We will also evaluate promoter methylation for genes associated with increased carcinogenic risk and the radiation response. Using these three strategies to evaluate tissue specific DNA methylation in irradiated mice we will be able to survey genome wide changes that have been linked to carcinogenic processes. As a whole, this project uses a systems biology approach that will provide information regarding tissue-specific effects of radiation LET, genetic factors, age, and post-irradiation time on miRNA-mediated gene regulation as it applies to the radiation response and radiation-induced carcinogenesis. This benefits not only our astronauts, but those of us on Earth, by increasing our understanding of the molecular mechanisms of cancer and the radiation response.
Task Progress:	During the past year we have published our first data set evaluating our panel of DNA methylation markers in C3H/HeN mice following irradiation using 600 MeV/n 56-Fe ions and 10, 30, and 100 cGy doses (5 cGy/min). Using quantitative pyrosequencing assays, we have evaluated the methylation status of the DAPK1, EVL, 14.3.3, p16, MGMT, and IGFBP3 genes, as well as the gene expression level for each locus. These genes were selected for their roles in such radiation responses or carcinogenic processes as apoptosis, metastasis, cell cycle regulation, or DNA repair. All of these loci generally have low levels of DNA methylation under normal circumstances, and aberrant changes in DNA methylation and gene expression have been associated with disease. We also evaluated two specific long interspersed nucleotide elements (LINE), the TF and A1 monomers, for radiation-induced changes in DNA methylation. These repeat elements typically are very highly methylated. The results of these experiments demonstrate clear A) tissue specific, B) time dependent, C) locus specific, and D) dose dependent epigenetic radiation responses. For the liver, no consistent changes in DNA methylation were observed. For the lung, we also observed no consistent changes in TF or A1 monomer DNA methylation. However, for the lung the remaining genes show clear trends, most of which were statistically significant. The changes in DNA methylation that we have observed for each gene oscillate with dose and time post exposure, with the largest effect observed in the 10 cGy group and at 1 and 7 days post irradiation. Given that all six genes generally show the same direction of change, it is possible that our observations are indicative of global methylation changes. A similar effect is not observed for the repeat elements, but these sequences are different in structure and regulation from the genes evaluated and they are highly methylated. It is possible that the oscillations that we observe for the genes represent a degree of overcorrection or instability as the cells work to regain homeostasis following exposure. That is not to say, though, that these are random, arbitrary changes. Rather the changes appear to represent a solid and specific biological response, the significance of which is not clear at this time. For most loci the inter-animal variability was quite small, and while there were instances of variation in the absolute percent methylation for a given CpG within a locus, whatever trend for radiation response was observed relative to control, it was generally reflected at all CpG. Evaluation of mRNA levels for these genes demonstrated that the changes in DNA methylation that we observed were not likely to alter gene expression. Since the TF and A1 repeat elements showed no radiation responsiveness in the above described study, we have developed an additional panel of pyrosequencing assays for intracisternal A particles (IAP) that have shown changes in DNA methylation following exposure to chemicals. These IAP are a different class of mouse retrotransposable elements than the LINE. IAPs are members of an endogenous retrovirus family that are also kept inactivated via high levels of DNA methylation. In addition to these new pyrosequencing assays, we have re-established and validated all of the other DNA methylation assays and have begun processing tissues archived from our previous NSRL campaigns. During the coming year we will complete the comparisons of DNA methylation changes based on age, genotype, and radiation quality. We are also in the process of isolating the nucleic acids in preparation for the qRT-PCR for gene expression and analysis of miR mediated changes in gene expression. We have also completed additional work characterizing cells that exhibit radiation induced genomic instability. In the first of two studies we tested the hypothesis that genomic instability and persistent oxidative stress observed in the chromosomally unstable progeny of irradiated cells contribute to unique DNA damage signatures

Task Progress:	characterized by altered levels of endogenous, potentially mutagenic, types of DNA damage and chromosomal breakage (Keszenman D.J. and Baulch J.E. DNA damage in cells exhibiting radiation induced genomic instability. Submitted Cancers). The measurement of oxypurine, oxypyrimidine, and abasic site endogenous DNA damage showed differences in non-DSB clusters among the four unstable clones evaluated as compared to irradiated but genomically stable clones and the parental cell line, and three of the four unstable clones had similar, increased levels of DSB clusters. Interestingly, the one unstable clone that did not show increased levels of DSB clusters was from a cell that had been exposed to high LET Fe ion radiation, while the other three had been exposed to low LET x-rays. The results of this study demonstrate that each unstable cell line has a unique spectrum of persistent damage and suggest that alterations in DNA damage signaling and repair may be related to the perpetuation of genomic instability. The manuscript describing this work has been prepared and submitted to the journal Cancers, where it is under review following first revision. The second study evaluated the same genomically unstable cells to determine whether epigenetic changes are among the mechanisms invoked in the perpetuation of the genomic instability phenotype. To do this, we measured DNA methylation, mRNA, and microRNA (miR) levels in the same chromosomally stable and unstable clonal cell lines used in the above mentioned study. These cells are clonally expanded single cell survivors of x- or Fe ion irradiation. While no changes in DNA methylation were observed for the gene promoters evaluated, increased LINE-1 methylation was observed for two unstable clones and decreased alu element methylation was observed for the other two unstable clones. These relationships also manifested for mRNA and miR expression. mRNA identified for the first set of clones were most similar to each other (261 mRNA), while the second set of clones were more similar to each other, and surprisingly also similar to the two stable clones (286 mRNA among these four clones). Pathway analysis showed enrichment for pathways involved in mitochondrial function and cellular redox, themes routinely invoked in genomic instability. The commonalities between the two subgroups of clones were also observed for miR. The number of miR for which anti-correlated mRNA were identified suggests that these miR exert functional effects in each clone. The results of this study demonstrate significant genetic and epigenetic changes in unstable cells, but similar changes almost equally common in chromosomally stable cells. Possible conclusions might be that the chromosomally stable clones have some other form of instability, or that some of the observed changes represent a sort of radiation signature and that other changes are related to genomic instability. Irrespective, these findings again suggested that a spectrum of changes both drive genomic instability and permit unstable cells to persist and proliferate. The manuscript describing this work has been prepared and submitted to the journal PLoSOne, where it is under review. The above-described studies have been presented in poster form at the 2013 Radiation Research Society meeting in New Orleans and at the 2014 NASA Space Radiation Investigators' Workshop in Galveston. I am an invited speaker at the 13th International Workshop on Radiation Damage to DNA. This meeting will be held in June of 2014 in Boston and my talk is titled 'Epigenetic responses to high and low LET radiations: effect of dose and radiation quality'. I am also an invited speaker at the 2014 Radiation Research Society meeting that will be held in Las Vegas in September of 2014.
Bibliography Type:	Description: (Last Updated: 05/01/2017)
Articles in Peer-reviewed Journals	Lima F, Ding D, Goetz W, Yang AJ, Baulch JE. "High LET 56Fe ion irradiation induces tissue-specific changes in DNA methylation in the mouse." Environmental and Molecular Mutagenesis. 2014 Apr;55(3):266-77. Epub 2013 Nov 30. http://dx.doi.org/10.1002/em.21832 ; PubMed PMID: 24723241 , Apr-2014